Chapter 3. Meromorphic Functions and the Logarithm

Zeros and poles, residues, singularities, the argument principle, homotopies, the complex logarithm

Contents

1 Zeros and poles **2**

2 The residue formula **3**

2.1 Examples 4

3 Singularities and meromorphic functions **5**

4 The argument principle and applications **7**

5 Homotopies and simply connected domains **8**

6 The complex logarithm **10**

7 Fourier series and harmonic functions **10**

Overview

Analytic functions are essentially characterized by their singularities (Riemann): holomorphic functions by their zeros, meromorphic ones by zeros and poles. We study removable singularities, poles and essential singularities; poles lead to the *calculus of residues*, built on $\int_C dz/z = 2\pi i$. The multivaluedness of $\int dz/z = \log z$ gives the *argument principle*, the open mapping theorem and the maximum principle; homotopies and simply connected domains define the logarithm.

1 Zeros and poles

A **point singularity** (**isolated singularity**) of f is a point z_0 where f is defined nearby but not at z_0. For $f(z) = z$ on the punctured plane, 0 is removable; $1/z$ grows to infinity at 0 (a pole); $e^{1/z}$ blows up along the positive reals, tends to 0 along the negative reals and oscillates boundedly along the imaginary axis.

z_0 is a **zero** of f if $f(z_0) = 0$. By analytic continuation, the zeros of a non-trivial holomorphic function are isolated.

Theorem 1.1

Suppose f is holomorphic in a connected open Ω, $f(z_0) = 0$, $f \not\equiv 0$. Then there are a neighborhood $U \subset \Omega$ of z_0, a non-vanishing holomorphic g on U and a unique $n \geq 1$ with $f(z) = (z - z_0)^n g(z)$ on U.

Proof. $f \not\equiv 0$ near z_0 (Theorem 4.8); with $f(z) = \sum_k a_k (z - z_0)^k$ near z_0 and n least with $a_n \neq 0$, $f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z)$, $g(z_0) = a_n \neq 0$.

Uniqueness: if also $f(z) = (z - z_0)^m h(z)$ with $h(z_0) \neq 0$ and $m > n$, then $g(z) = (z - z_0)^{m-n} h(z)$ and letting $z \to z_0$ gives $g(z_0) = 0$, a contradiction; $m < n$ similarly forces $h(z_0) = 0$. So $m = n$ and $h = g$. ∎

f has a **zero of order** n (multiplicity n) at z_0; a zero of order 1 is **simple**. A **deleted neighborhood** of z_0 is $\{z : 0 < |z - z_0| < r\}$. A function f defined in a deleted neighborhood of z_0 has a **pole** at z_0 if $1/f$, defined to be zero at z_0, is holomorphic in a full neighborhood of z_0.

Theorem 1.2

If f has a pole at $z_0 \in \Omega$, then near z_0 there are a non-vanishing holomorphic h and a unique $n \geq 1$ with $f(z) = (z - z_0)^{-n} h(z)$.

Proof. By Theorem 1.1, $1/f(z) = (z - z_0)^n g(z)$ with g holomorphic and non-vanishing near z_0; take $h = 1/g$. ∎

n is the **order** of the pole (it describes the rate of growth); a pole of order 1 is **simple**.

Theorem 1.3

If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0} + G(z), \tag{1}$$

where G is a holomorphic function in a neighborhood of z_0.

Proof. Expand $h(z) = A_0 + A_1(z - z_0) + \cdots$ in Theorem 1.2; then $f(z) = (z - z_0)^{-n}(A_0 + A_1(z - z_0) + \cdots)$ has the stated form. ∎

$\frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-1}}{z-z_0}$ is the **principal part** of f at z_0, and $\operatorname{res}_{z_0} f = a_{-1}$ the **residue**. The other terms have primitives near z_0, so $\frac{1}{2\pi i}\int_C P(z)\,dz = a_{-1}$ for the principal part P and any circle C about z_0. For a simple pole, $\operatorname{res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f(z)$.

Theorem 1.4

If f has a pole of order n at z_0, then

$$\operatorname{res}_{z_0} f = \lim_{z\to z_0} \frac{1}{(n-1)!}\Big(\frac{d}{dz}\Big)^{n-1}(z-z_0)^n f(z).$$

Proof. By (1), $(z-z_0)^n f(z) = a_{-n}+a_{-n+1}(z-z_0)+\cdots+a_{-1}(z-z_0)^{n-1}+G(z)(z-z_0)^n$; differentiate $n-1$ times and let $z\to z_0$. ∎

2 The residue formula

Motivation. Each singularity inside a contour contributes its residue, turning hard real integrals into a few coefficients.

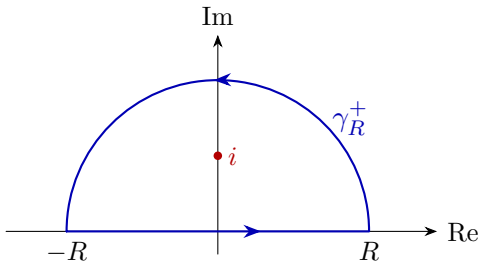


Figure. *The standard semicircular contour for* $\int_{-\infty}^{\infty}\frac{dx}{1+x^2}$*: only the pole at* i *lies inside, the arc contributes* $O(1/R)\to 0$*, and the residue formula gives* $2\pi i\cdot\frac{1}{2i}=\pi$*.*

Theorem 2.1

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then

$$\int_C f(z)\,dz = 2\pi i \operatorname{res}_{z_0} f.$$

Proof. A keyhole avoiding the pole, with corridor width $\to 0$, gives $\int_C f = \int_{C_\epsilon} f$ for the circle C_ϵ of radius ϵ about z_0. By the Cauchy integral formula $\frac{1}{2\pi i}\int_{C_\epsilon}\frac{a_{-1}}{z-z_0}\,dz = a_{-1}$; by Corollary 4.2, $\int_{C_\epsilon}\frac{a_{-k}}{(z-z_0)^k}\,dz = 0$ for $k>1$; $\int_{C_\epsilon} G = 0$ by Cauchy. By (1), $\int_{C_\epsilon} f = 2\pi i a_{-1}$. ∎

Corollary 2.2

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1,\dots,z_N$ inside C. Then

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Proof. A multiple keyhole with a loop around each pole reduces $\int_C$ to a sum over small circles; apply Theorem 2.1. ∎

Corollary 2.3

If f is holomorphic in an open set containing a toy contour γ and its interior, except for poles $z_1, \dots, z_N$ inside γ, then

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Here γ has positive orientation.

(Choose a keyhole appropriate to the toy contour and reduce to small circles around the poles.) This identity is the **residue formula**.

2.1 Examples

To compute $\int_{-\infty}^{\infty} f(x)\,dx$, choose toy contours γ_R with $\int_{\gamma_R} f \to \int_{-\infty}^{\infty} f(x)\,dx$ and evaluate $\int_{\gamma_R} f$ by residues; the decay of f guides the choice.

Example 1

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi. \tag{2}$$

(Equivalently $\int P_y(x)\,dx = 1$ for the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$.) $f(z) = 1/((z-i)(z+i))$ has simple poles at $\pm i$, $\operatorname{res}_i f = 1/2i$. With $\gamma_R = [-R, R] \cup C_R^+$, $\int_{\gamma_R} f = \pi$ for large R, and $|f| \le B/|z|^2$ on C_R^+ gives $\left|\int_{C_R^+} f\right| \le \frac{B}{R^2}\pi R \to 0$. So $\int_{-\infty}^{\infty} f(x)\,dx = \pi$.

Example 2

An integral that plays an important role in Chapter 6:

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x}\,dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1.$$

Let $f(z) = e^{az}/(1+e^z)$ and γ_R the rectangle with sides on $\mathbb{R}$ and on $\operatorname{Im}(z) = 2\pi$, vertices $\pm R$, $\pm R + 2\pi i$. The only zero of the denominator inside is $z = \pi i$. There

$$(z-\pi i) f(z) = e^{az}\frac{z-\pi i}{e^z - e^{\pi i}}, \qquad \lim_{z\to \pi i} \frac{e^z - e^{\pi i}}{z - \pi i} = e^{\pi i} = -1,$$

so f has a simple pole at πi with residue $-e^{a\pi i}$, and

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

Let $I_R = \int_{-R}^{R} f(x)\,dx \to I$. The top side contributes $-e^{2\pi i a} I_R$. On $A_R = \{R + it : 0 \le t \le 2\pi\}$, $\left|\int_{A_R} f\right| \le \int_0^{2\pi} \left|\frac{e^{a(R+it)}}{1+e^{R+it}}\right| dt \le C e^{(a-1)R} \to 0$ $(a < 1)$; the left side is $\le C e^{-aR} \to 0$ $(a > 0)$. So (3) gives $I - e^{2\pi i a} I = -2\pi i e^{a\pi i}$, whence

$$I = -2\pi i \frac{e^{a\pi i}}{1 - e^{2\pi i a}} = \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} = \frac{\pi}{\sin \pi a}.$$

Example 3

$1/\cosh \pi x$ is its own Fourier transform: with $\cosh z = \frac{e^z+e^{-z}}{2}$,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{\cosh \pi x}\, dx = \frac{1}{\cosh \pi \xi}.$$

Let $f(z) = e^{-2\pi i z\xi}/\cosh \pi z$ and γ_R the rectangle with vertices $\pm R$, $\pm R+2i$. The denominator vanishes when $e^{2\pi z} = -1$; inside the rectangle the poles are $\alpha = i/2$ and $\beta = 3i/2$. Since

$$(z-\alpha)f(z) = 2e^{-2\pi i z\xi} e^{\pi z} \frac{z-\alpha}{e^{2\pi z} - e^{2\pi \alpha}}, \qquad \lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i \alpha\xi} e^{\pi\alpha} \frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

f has simple poles at α, β with residues $e^{\pi\xi}/(\pi i)$, $-e^{3\pi\xi}/(\pi i)$. On $z = R + iy$, $0 \le y \le 2$: $|e^{-2\pi i z\xi}| \le e^{4\pi|\xi|}$ and $|\cosh \pi z| \ge \frac{1}{2}(e^{\pi R} - e^{-\pi R}) \to \infty$, so the vertical sides vanish in the limit. As $\cosh \pi z$ has period $2i$, the top side gives $-e^{4\pi\xi} I$. The residue formula gives

$$I - e^{4\pi\xi} I = 2\pi i\Big(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i}\Big) = -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}),$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$,

$$I = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}.$$

A similar argument gives, for $0 < a < 1$, with $\sinh z = (e^z - e^{-z})/2$,

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi} \frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\, dx = \frac{2\sinh 2\pi a\xi}{\sinh 2\pi\xi};$$

$a = 1/2$ is the case above. This yields the Poisson kernel for the strip, and is used in the two-squares theorem (Chapter 10).

3 Singularities and meromorphic functions

Motivation. Removable singularities, poles and essential singularities behave very differently; near an essential one f comes close to every value.

Let f be holomorphic in Ω except possibly at $z_0 \in \Omega$. If f can be defined at z_0 so that it becomes holomorphic in all of Ω, z_0 is a **removable singularity**.

Theorem 3.1 (Riemann's theorem on removable singularities)

Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω. If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.

Proof. Take a small disc D centered at z_0 with $\overline{D} \subset \Omega$, boundary circle C positively oriented. We show that for $z \in D$, $z \ne z_0$,

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta. \tag{4}$$

By Theorem 5.4 the right side is holomorphic in D and equals f for $z \ne z_0$. To prove (4), fix z and shrink the corridors of a multiple keyhole avoiding z and z_0:

$$\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = 0,$$

with γ_ϵ, γ'_ϵ negatively oriented circles of radius ϵ about z, z_0. As for the Cauchy formula, $\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = -2\pi i f(z)$; f is bounded and γ'_ϵ stays away from z, so $|\int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta| \le C\epsilon$. Let $\epsilon \to 0$. ■

Corollary 3.2

Suppose that f has an isolated singularity at the point z_0. Then z_0 is a pole of f if and only if $|f(z)| \to \infty$ as $z \to z_0$.

Proof. If z_0 is a pole, $1/f$ has a zero there, so $|f| \to \infty$. Conversely, if $|f| \to \infty$, $1/f$ is bounded near z_0 with $1/f \to 0$, so it has a removable singularity with value 0: a pole. ■

Isolated singularities are **removable** (f bounded near z_0), **poles** ($|f| \to \infty$), or **essential** (all others, e.g. $e^{1/z}$ at 0), where f behaves erratically:

Theorem 3.3 (Casorati–Weierstrass)

If f is holomorphic in $D_r(z_0) - \{z_0\}$ with an essential singularity at z_0, then $f(D_r(z_0) - \{z_0\})$ is dense in $\mathbb{C}$.

Proof. Otherwise $|f(z) - w| > \delta$ on the punctured disc for some w, $\delta > 0$. Then $g = 1/(f - w)$ is bounded by $1/\delta$, so removable at z_0 (Theorem 3.1). If $g(z_0) \neq 0$, $f - w = 1/g$ is holomorphic at z_0; if $g(z_0) = 0$, $f - w$ has a pole. Either contradicts essentiality. ■

Picard proved much more: under these hypotheses f takes every complex value infinitely many times with at most one exception.

Definition (meromorphic function)

f on an open Ω is **meromorphic** if for a sequence $\{z_0, z_1, \dots\}$ with no limit points in Ω, (i) f is holomorphic in $\Omega - \{z_0, z_1, \dots\}$, and (ii) f has poles at the z_k.

If f is holomorphic for large z, let $F(z) = f(1/z)$. f has a **pole**, **essential singularity** or **removable singularity at infinity** if F has one at 0. A meromorphic f on $\mathbb{C}$ that is holomorphic or has a pole at infinity is **meromorphic in the extended complex plane**. The opening principle in its simplest form:

Theorem 3.4

The meromorphic functions in the extended complex plane are the rational functions.

Proof. $f(1/z)$ is holomorphic in a deleted neighborhood of 0, so f has finitely many poles $z_1, \dots, z_n$. Near z_k, $f = f_k + g_k$ with principal part f_k (a polynomial in $1/(z - z_k)$) and g_k holomorphic. Likewise $f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$ with $\tilde{f}_\infty$ a polynomial in $1/z$; $f_\infty(z) = \tilde{f}_\infty(1/z)$ is a polynomial. $H = f - f_\infty - \sum_k f_k$ is entire and bounded ($H(1/z)$ is bounded near 0), hence constant by Liouville; so f is rational. ■

Consequently, a rational function is determined up to a multiplicative constant by the locations and multiplicities of its zeros and poles.

The Riemann sphere. In $\mathbb{R}^3$ identify the XY-plane with $\mathbb{C}$, and let $\mathbb{S}$ be the sphere of radius $1/2$ centered at $(0, 0, 1/2)$, with north pole $N = (0, 0, 1)$. For $W \in \mathbb{S} - \{N\}$, the line NW meets the plane in $w = x + iy$, the **stereographic projection** of W; this bijection $\mathbb{S} - \{N\} \to \mathbb{C}$ is

$$x = \frac{X}{1 - Z}, \quad y = \frac{Y}{1 - Z}; \qquad X = \frac{x}{x^2 + y^2 + 1}, \quad Y = \frac{y}{x^2 + y^2 + 1}, \quad Z = \frac{x^2 + y^2}{x^2 + y^2 + 1}.$$

As $|w| \to \infty$, $W \to N$, the "point at infinity". The extended plane is the **Riemann sphere** $\mathbb{S}$ (*one-point compactification* of $\mathbb{C}$), and a meromorphic function on it maps $\mathbb{S}$ to itself, poles to N.

4 The argument principle and applications

$\log f(z) = \log|f(z)| + i \arg f(z)$ is multiple-valued, but f'/f is single-valued, and $\int_\gamma \frac{f'(z)}{f(z)}\,dz$ is the change of $\arg f$ along γ, determined by the zeros and poles inside. Logarithmic derivatives are additive:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}, \qquad \frac{(\prod_{k=1}^N f_k)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

At a zero of order n, $f = (z-z_0)^n g$ with $g \neq 0$, so $\frac{f'}{f} = \frac{n}{z-z_0} + G$, $G = g'/g$ holomorphic: a simple pole with residue n. At a pole of order n, $f = (z-z_0)^{-n} h$ and $\frac{f'}{f} = \frac{-n}{z-z_0} + H$. The residue formula gives:

Theorem 4.1 (Argument principle)

Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C, then

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,dz = (\text{number of zeros of } f \text{ inside } C) - (\text{number of poles of } f \text{ inside } C),$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2

The above theorem holds for toy contours.

Three applications: Rouché's theorem (a continuity statement), the open mapping theorem, and the maximum modulus principle.

Theorem 4.3 (Rouché's theorem)

If f, g are holomorphic in an open set containing a circle C and its interior and $|f| > |g|$ on C, then f and $f+g$ have the same number of zeros inside C.

Proof. Let $f_t = f + tg$, $t \in [0,1]$, and n_t its number of zeros inside C. $|f| > |g|$ on C means $f_t \neq 0$ on C, so $n_t = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\,dz$ by the argument principle. The integrand is jointly continuous in (t,z), so n_t is a continuous integer-valued function on $[0,1]$, hence constant: $n_0 = n_1$. ∎

A mapping is **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem)

If f is holomorphic and non-constant in a region Ω, then f is open.

Proof. Let $w_0 = f(z_0)$; we show all w near w_0 are values. Write $f - w = (f - w_0) + (w_0 - w) = F + G$. Choose $\delta > 0$ with $\overline{D_\delta(z_0)} \subset \Omega$ and $f \neq w_0$ on $|z - z_0| = \delta$ (isolated zeros), and $\epsilon > 0$ with $|f - w_0| \geq \epsilon$ there. For $|w - w_0| < \epsilon$, $|F| > |G|$ on the circle, and Rouché gives $F + G$ a zero inside, as F has one (z_0). ∎

The maximum of f in Ω means the maximum of $|f|$ in Ω.

Theorem 4.5 (Maximum modulus principle)

If f is a non-constant holomorphic function in a region Ω, then f cannot attain a maximum in Ω.

Proof. If $|f|$ had a maximum at z_0, the open set $f(D)$, D a small disc about z_0, would contain points of modulus $> |f(z_0)|$. ■

Corollary 4.6

Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then

$$\sup_{z\in\Omega}|f(z)| \leq \sup_{z\in\overline{\Omega}-\Omega}|f(z)|.$$

Proof. $|f|$ attains its maximum on the compact set $\overline{\Omega}$, and not in Ω if f is non-constant; if f is constant the conclusion is trivial. ■

Remark

Boundedness of Ω is essential: on the open first quadrant, $F(z) = e^{-iz^2}$ is entire with $|F| = 1$ on both boundary half-lines, yet $F(re^{i\pi/4}) = e^{r^2}$ is unbounded.

5 Homotopies and simply connected domains

In which regions does a holomorphic function have a primitive? This global question underlies the general Cauchy theorem and the logarithm.

Definition (homotopic curves)

Curves $\gamma_0, \gamma_1 \subset \Omega$ on $[a, b]$ with common end-points α, β are **homotopic in** Ω if there are curves $\gamma_s \subset \Omega$, $0 \leq s \leq 1$, on $[a, b]$ with $\gamma_s(a) = \alpha$, $\gamma_s(b) = \beta$, equal to γ_0, γ_1 at $s = 0, 1$, and $\gamma_s(t)$ jointly continuous in $(s, t) \in [0, 1] \times [a, b]$.

Loosely, one curve can be deformed into the other continuously without leaving Ω.

Theorem 5.1

If f is holomorphic in Ω, then $\int_{\gamma_0} f(z)\,dz = \int_{\gamma_1} f(z)\,dz$ whenever the two curves γ_0 and γ_1 are homotopic in Ω.

Proof. Close curves with the same end-points give the same integral. $F(s, t) = \gamma_s(t)$ has compact image K. Some $\epsilon > 0$ makes every disc of radius 3ϵ about a point of K lie in Ω: otherwise $z_\ell \in K$, $w_\ell \in \Omega^c$ with $|z_\ell - w_\ell| < 1/\ell$ have a subsequence $z_{\ell_k}, w_{\ell_k} \to z \in K \cap \Omega^c$ (Ω^c closed), a contradiction. By uniform continuity choose δ with $\sup_t|\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon$ for $|s_1 - s_2| < \delta$.

Fix such s_1, s_2. Choose discs $D_0, \ldots, D_n$ of radius 2ϵ covering both curves and consecutive points $z_0, \ldots, z_{n+1}$ on γ_{s_1}, $w_0, \ldots, w_{n+1}$ on γ_{s_2} with $z_i, z_{i+1}, w_i, w_{i+1} \in D_i$, $z_0 = w_0$, $z_{n+1} = w_{n+1}$ the common end-points. Let F_i be a primitive of f on D_i (Theorem 2.1). On $D_i \cap D_{i+1}$, $F_i - F_{i+1}$ is constant, so $F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1})$, i.e.

$$F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}). \tag{5}$$

Hence

$$\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f = \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n} [F_i(w_{i+1}) - F_i(w_i)] = \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i))],$$

which telescopes by (5) to $F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) = 0$. Subdividing $[0,1]$ into intervals of length $< \delta$, finitely many applications take us from γ_0 to γ_1. ■

Definition (simply connected)

A region Ω is **simply connected** if any two curves in Ω with the same end-points are homotopic.

Example 1

A disc is simply connected: $\gamma_s(t) = (1-s)\gamma_0(t) + s\gamma_1(t)$ stays in D. So are rectangles and all open convex sets.

Example 2

The slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ is simply connected: write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ with $r_j > 0$ and θ_j continuous, $|\theta_j(t)| < \pi$, and let $\gamma_s(t) = r_s(t)e^{i\theta_s(t)}$ with $r_s = (1-s)r_0 + sr_1$, $\theta_s = (1-s)\theta_0 + s\theta_1$; then $\gamma_s(t) \in \Omega$.

Example 3

With some effort one shows that the interior of a toy contour is simply connected, by dividing it into several subregions.

Example 4

The punctured plane $\mathbb{C} - \{0\}$ is not simply connected (proved after Corollary 5.3).

Theorem 5.2

Any holomorphic function in a simply connected domain has a primitive.

Proof. Fix $z_0 \in \Omega$ and let $F(z) = \int_\gamma f(w)\,dw$ over any curve $\gamma \subset \Omega$ from z_0 to z (independent of γ by Theorem 5.1). Then $F(z+h) - F(z) = \int_\eta f$ over the segment η from z to $z+h$, and as in Theorem 2.1, $F'(z) = f(z)$. ■

Corollary 5.3

If f is holomorphic in the simply connected region Ω, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in Ω.

That $\mathbb{C} - \{0\}$ is not simply connected now follows rigorously since $\int_C dz/z = 2\pi i \neq 0$ over the unit circle.

6 The complex logarithm

For $z = re^{i\theta}$ one wants $\log z = \log r + i\theta$, but θ is only defined mod 2π: a continuous choice exists locally, not globally (winding once around 0 adds $2\pi i$). A single-valued logarithm needs a restricted domain: a **branch** (**sheet**).

Theorem 6.1

Suppose that Ω is simply connected with $1 \in \Omega$, and $0 \notin \Omega$. Then in Ω there is a branch of the logarithm $F(z) = \log_\Omega(z)$ so that (i) F is holomorphic in Ω, (ii) $e^{F(z)} = z$ for all $z \in \Omega$, (iii) $F(r) = \log r$ whenever r is a real number and near 1.

So each branch $\log_\Omega(z)$ extends the standard logarithm.

Proof. Let $\log_\Omega(z) = F(z) = \int_\gamma \frac{dw}{w}$, $\gamma \subset \Omega$ from 1 to z (path-independent). As in Theorem 5.2, $F'(z) = 1/z$: (i). For (ii), $\frac{d}{dz}(ze^{-F(z)}) = (1 - zF'(z))e^{-F(z)} = 0$, so $ze^{-F(z)}$ is constant (Corollary 3.4), equal to 1 at $z = 1$. For (iii), integrate along the real segment: $F(r) = \int_1^r \frac{dx}{x} = \log r$. ■

In the slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ we have the **principal branch** $\log z = \log r + i\theta$ for $z = re^{i\theta}$, $|\theta| < \pi$: integrate along the segment from 1 to r and the arc from r to z,

$$\log z = \int_1^r \frac{dx}{x} + \int_{\text{arc}} \frac{dw}{w} = \log r + \int_0^\theta \frac{ire^{it}}{re^{it}}\, dt = \log r + i\theta.$$

In general $\log(z_1 z_2) \neq \log z_1 + \log z_2$: for $z_1 = z_2 = e^{2\pi i/3}$, $\log z_1 = \log z_2 = 2\pi i/3$, while $z_1 z_2 = e^{-2\pi i/3}$ and $\log(z_1 z_2) = -2\pi i/3$. For the principal branch,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = -\sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} \qquad \text{for } |z| < 1, \tag{6}$$

since both sides have derivative $1/(1+z)$ and vanish at $z = 0$.

For Ω simply connected with $1 \in \Omega$, $0 \notin \Omega$, and $\alpha \in \mathbb{C}$, define $z^\alpha = e^{\alpha \log z}$ (branch with $\log 1 = 0$). Then $1^\alpha = 1$, and for $\alpha = 1/n$, $(z^{1/n})^n = \prod_{k=1}^n e^{\frac{1}{n}\log z} = e^{\log z} = z$.

Theorem 6.2

If f is a nowhere vanishing holomorphic function in a simply connected region Ω, then there exists a holomorphic function g on Ω such that $f(z) = e^{g(z)}$.

g may be denoted $\log f(z)$; it determines a "branch" of that logarithm.

Proof. Fix $z_0 \in \Omega$ and let $g(z) = \int_\gamma \frac{f'(w)}{f(w)}\, dw + c_0$ over a path from z_0 to z, with $e^{c_0} = f(z_0)$. As in Theorem 2.1, $g' = f'/f$, so $\frac{d}{dz}(fe^{-g}) = 0$ and $fe^{-g} \equiv f(z_0)e^{-c_0} = 1$. ■

7 Fourier series and harmonic functions

Motivation. On a circle a power series is a Fourier series; this gives the mean value property for harmonic functions.

Let f be holomorphic in $D_R(z_0)$, $f(z) = \sum_{n\geq 0} a_n (z - z_0)^n$.

Theorem 7.1

The coefficients of the power series expansion of f are given by

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula over the circle γ of radius r about z_0 gives, with $\zeta = z_0 + re^{i\theta}$,

$$a_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(re^{i\theta})^{n+1}} rie^{i\theta}\, d\theta = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta.$$

For $n < 0$ the same computation gives $\frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta$, and since $-n > 0$, $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc; by Cauchy's theorem the integral vanishes. ■

So on a circle the Fourier coefficients of f vanish for $n < 0$ and equal $a_n r^n$ for $n \geq 0$. Since $a_0 = f(z_0)$:

Corollary 7.2 (Mean-value property)

If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

Corollary 7.3

If f is holomorphic in a disc $D_R(z_0)$, and $u = \mathrm{Re}(f)$, then

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

Take real parts: every harmonic u in a disc is the real part of a holomorphic function there.